

# Dongqi Lin, Ph. D.

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## RESEARCH PROFILE

I am an atmospheric scientist studying how atmospheric processes interact across scales, from turbulence to convective storms to climate. By linking processes that are usually studied in isolation, and by combining high-resolution modelling with observations, my research centres on the physical understanding of high-impact weather: how it shapes, and is shaped by, Earth's climate.

## EDUCATION

- Apr. 2019 – Ph. D. in Physics, University of Canterbury (UC), New Zealand  
Oct. 2022
- Sep. 2017 – MRes in Climate and Atmospheric Science, University of Leeds, United Kingdom  
Oct. 2018 (Distinction)
- Sep. 2016 – BSc in Applied Mathematics and Meteorology, University of Reading, United Kingdom  
Jun. 2017 (1st Class Honours)
- Sep. 2013 – BSc in Mathematics and Applied Mathematics, Nanjing University of Information Science  
Jun. 2017 and Technology (NUIST), China (average grade of 87%)

## PROFESSIONAL APPOINTMENTS

- Apr. 2024 – Research Fellow, the ARC Centre of Excellence for Climate Extremes (CLEX), Monash  
Present University, Melbourne, VIC, Australia
- Aug. 2022 – Postdoctoral Fellow, University of Canterbury, Christchurch, New Zealand  
Mar. 2024

## PUBLICATIONS

- Lin, D., Saunders, K.R. and Singh, M.S.: Mapping the Spatial Scales of Australian Extreme Precipitation Using Daily Rain Gauges. *International Journal of Climatology*, p.e70283. 10.1002/joc.70283, 2026.
- Lin, D., Katurji, M., Zawar-Reza, P., Malyarenko, A., Valencia, A. and Zhang, J.: High-fidelity experimental simulations of atmospheric flow near low-intensity fires in a wildland-urban interface. *Journal of Geophysical Research: Atmospheres*, 130(19), p.e2025JD043641. 10.1029/2025JD043641, 2025.
- Valencia A., Penney G., Baker G., Gorham D., Gill F., Lin, D., Katurji M., and Power A.: Development of wildland fire exposure indicator for performance-based design applications. iCFSERP20230170, 2024. [Peer-reviewed conference proceeding]
- Lin, D., Zhang, J., Khan, B., Katurji, M. and Revell, L.E.: GEO4PALM v1.1: an open-source geospatial data processing toolkit for the PALM model system. *Geoscientific Model Development*, 17, 815-845, 10.5194/gmd-17-815-2024, 2024.
- Bhatti, Y. A., Revell, L. E., McDonald, A. J., Archibald, A. T., Schuddeboom, A. J., Williams, J., Hardacre, C., Mulcahy, J., and Lin, D.: Aerosol and Dimethyl Sulfide Sensitivity to Global Sulfate Chemistry Schemes, *Journal of Geophysical Research: Atmospheres*, 129, e2023JD040635. 10.1029/2023JD040635, 2024.
- Lin, D., Katurji, M., Revell, L. E., Khan, B., and Sturman, A.: Investigating multiscale meteorological controls and impact of soil moisture heterogeneity on radiation fog in complex terrain using semi-idealised simulations, *Atmospheric Chemistry and Physics*, 23, 14451-14479, 10.5194/acp-23-14451-2023, 2023.
- Lin, D., Katurji, M., Revell, L. E., Khan, B., Osborne, N., Soltanzadeh, I., and Kremser, S.: Fog type classification using a modified Richardson number for Christchurch, New Zealand. *International Journal of Climatology*, 43(1), 314-330. 10.1002/joc.7761, 2023.
- Nathan, B., Kremser, S., Mikaloff-Fletcher, S., Bodeker, G., Bird, L., Dale, E., Lin, D., Olivares, G., and Somervell, E.: The MAPM (Mapping Air Pollution eMissions) method for inferring particulate matter emissions maps at city scale from in situ concentration measurements: description and demonstration of capability, *Atmospheric Chemistry and Physics*, 21, 14089-14108, 10.5194/acp-21-14089-2021, 2021.
- Lin, D., Khan, B., Katurji, M., Bird, L., Faria, R., and Revell, L. E.: WRF4PALM v1.0: a mesoscale dynamical driver for the microscale PALM model system 6.0, *Geoscientific Model Development*, 14, 2503-2524, 10.5194/gmd-14-2503-2021, 2021.

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**Lin, D.**, Pickering, B., and Neely III, R. R.: Relating the Radar Bright Band and Its Strength to Surface Rainfall Rate Using an Automated Approach, *Journal of Hydrometeorology*, 21(2), 335-353. 10.1175/JHM-D-19-0085.1, 2020.

**Manuscript in review**

**Lin, D.**, Singh, M. S., Li, Y.-X., and Bowden, A. J.: Downpours and deluges: Understanding the contrasting environments for localised and widespread heavy rainfall in the tropics, EGU sphere; Weather and Climate Dynamics [preprint], 10.5194/egusphere-2026-4291, 2026.

Pletzer, T., Pletzer, A., Andrews, T. C., **Lin, D.**, and Katurji, M.: Mimetic interpolation for finite-time Lyapunov exponent computation near irregular domain boundaries, EGU sphere; Geoscientific Model Development [preprint], 10.5194/egusphere-2026-3377, 2026.

## AWARDS AND GRANTS

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Jun. 2025 Advancing Diversity in Science Grant (AUD 10,000), Faculty of Science, Monash University

Sep. 2024 IOP Publishing Trusted Reviewer

Oct. 2023 UC Library Open Access Fund (NZD 4,500)

2019 – 2022 UC Connect doctoral scholarship (NZD 21,000 stipend + fees) collaborating with the MBIE project: Mapping Air Pollution eMissions

2021 – 2022 UC Foundation Doctoral Publication Prize

Nov. 2021 Best presentation (3rd year student), School of Physical and Chemical Sciences Postgraduate Annual Symposium, University of Canterbury

Nov. 2020 Best student talk, New Zealand MetSoc Annual Conference

Jul. 2014 Scholarship for Geographic Information System (GIS) training (CNY 20,000), Saint Louis University, St. Louis, Missouri, United States

2013 – 2017 The Third-Class Scholarship awarded by NUIST, Nanjing, China

## CONFERENCES AND INVITED TALKS

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**Lin, D.**, Singh, M. S., Li, Y.-X., and Bowden, A. J.: Two ways to rain hard: mean versus local peak intensity in tropical convection, Asia Oceania Geosciences Society (AOGS), Fukuoka, Japan, 2026.

Pletzer, T., **Lin, D.**, Zawar-Reza, P., and Katurji, M.: Revealing Hidden Coherent Structures in Plume Atmospheric Dynamics, The 37th international conference on Alpine Meteorology, Croatia, 2025.

**Lin, D.**, Saunders, K. R., and Singh, M. S.: Do the environments for large-scale and small-scale precipitation extremes differ? The 31st conference of Australian Meteorological and Oceanographic Society (AMOS), Cairns, Australia, 2025. Also at Climate and Ocean: Variability, Predictability and Change (CLIVAR), Lorne, Australia, 2025.

**Lin, D.**, Katurji, M., Zhang, J., Malyarenko, A., and Valencia A.: Fire-atmosphere dynamics in a rural-urban interface using turbulence resolving simulations. AGU Fall meeting, San Francisco, 2023.

Le Cap, C., Carlier, J., Katurji, M., **Lin, D.**, Quénot, H., Georgeault, P., Buisson, E., and Heitz, D.: Towards numerical simulation of a wind machine during spring frost calibrated with field measurements, EGU General Assembly 2023, Vienna, Austria, 2023.

**Lin, D.**, Katurji, M., and Purdie, H.: Impact of crevasses on surface energy balance at an alpine glacier, EGU General Assembly 2023, Vienna, Austria, 2023.

**Lin, D.**, Katurji, M., Valencia, A., and Gill, F.: Fire-atmosphere dynamics at a rural-urban interface using turbulence-resolving meteorological simulations, EGU General Assembly 2023, Vienna, Austria, 2023.

Katurji, M., Zhang, J., Valencia, A., **Lin, D.**, Strand, T., Pearce, G., Finney, M., Clements, C., and Gross, S.: Atmosphere and fire interactions from the New Zealand experimental burn campaigns, EGU General Assembly 2023, Vienna, Austria, 2023.

Valencia, A., Katurji, M., **Lin, D.**, Gross, S., Zhang, J., Pearce, G., Finney, M., and Strand, T.: Unravelling wildland fire morphology and structure using image velocimetry, EGU General Assembly 2023, Vienna, Austria, 2023.

**Lin, D.**, Weather Research and Forecasting (WRF) model for air pollution applications, Clean Air Society of Australia and New Zealand (CASANZ) workshop, online, 2023. [Invited talk]

**Lin, D.**, Katurji, M., Khan, B., Revell, L.E.: Fog dynamics and predictability using a coupled urban-atmosphere numerical model for Christchurch, New Zealand, 13th International Conference on Southern Hemisphere Meteorology and Oceanography (ICSHMO), online, 2022.

**Lin, D.**, Khan, B., Katurji, M., Bird, L., Faria, R., and Revell, L.: WRF4PALM v1.0: A Mesoscale Dynamic Driver for the Microscale PALM Model System 6.0, EGU General Assembly 2021, online, 2021.

Bowery, A., Sparrow, S., Wallom, D., Carver, G., Köhler, M., Weisheimer, A., **Lin, D.**, Katurji, M., and Rosier, S.: Expanding the climateprediction.net project: the creation of a multi model distributed computing infrastructure for climate science. Academia Sinica Grid Computing Centre (ASGC), online, 2021.

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**Lin, D.,** Pickering, B., and Neely III, R. R.: Identifying the Microphysical Difference in UK Precipitation for Periods with and without Radar Bright Band. National Facilities for Atmospheric and Radar Research (NFARR) User Meeting, Chilbolton, United Kingdom, 2018. [Invited talk]

## TEACHING AND SUPERVISION

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- Co-supervisor. Master of Business Analytics (MBAt) internship projects ETC5543. Supervised two MBAt students on geospatial metadata and automatic siting classification for weather stations in the Netherlands, collaborating with Koninklijk Nederlands Meteorologisch Instituut (KNMI). (Monash University, 2026)
- Guest lecturer. EAE5053 Advanced Spatial Analysis and Modelling. Principles and practice of wildfire modelling. (Monash University, Sep. 2025)
- Supervisor, Summer research project. Supervised an undergraduate student investigating discrepancies in radar-derived precipitation extremes. (Monash University, Feb. – Apr. 2025)
- Guest lecturer. GEOG310 Weather Systems. Fog and its weather systems. (UC, Oct. 2023)
- Co-supervisor. Undergraduate research projects. Supervised one student assessing the impacts of the 2019–2020 Australian bushfires on the Tasman Glacier, and another student developing 3D visualisation of Doppler wind lidar observations. (UC, Aug. – Oct. 2023)
- Guest lecturer, course design, and teaching assistant. PHYS330 Climate Modelling. (UC, 2020–2022)
- Organiser. Leading and teaching a two-day in-person WRF workshop for the New Zealand MetSoc. (Nov. 2020)

## RESEARCH CONSULTANCIES

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- Kea Aerospace: Observations for solar-powered stratospheric aircraft operations (2023).
- Fire and Emergency New Zealand (FENZ): Extreme wind investigation for the 2020 Lake Ōhau fire (2023).
- FENZ: Fire weather index and future fire risk projection in New Zealand (2022).
- Antarctica New Zealand: Ash hazard and impact for the Scott Base redevelopment (2021).
- Pattle Delamore Partners: air pollution modelling for Whangarei, New Zealand (2021).
- DLA Piper New Zealand: WRF simulations of Early Valley Road wind conditions during the Port Hills 2017 fires (2020).

## RESEARCH VISITS AND COLLABORATIONS

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- Max Planck Institute for Meteorology, Dr Hans Segura. Collaborative work on global storm-resolving models. (Dec. 2025)
- The Arctic University of Norway (UiT), Dr Nikolas Aksamit. Collaborative work on Lagrangian coherent structures in tropical convection. (Nov. 2025)
- Karlsruhe Institute of Technology (KIT), Dr Basit Khan. Collaborative work on dynamical downscaling for the WRF and PALM modelling systems. (Aug. – Sep. 2019)

## SERVICE AND LEADERSHIP

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| <i>Ongoing</i>               | Reviewer for top journals including Geoscientific Model Development, JGR: Atmospheres, Geophysical Research Letters, Environmental Research Letters, Atmospheric Chemistry and Physics, Urban Environments, and AMS journals. |
| <i>Jan. 2026 – present</i>   | Early Career Researcher (ECR) Committee, Faculty of Science, Monash University.                                                                                                                                               |
| <i>Jun. 2025 – present</i>   | ECR Committee, School of Earth Atmosphere and Environment, Monash University.                                                                                                                                                 |
| <i>Jul. 2024 – Mar. 2025</i> | ECR Committee, CLEX.                                                                                                                                                                                                          |
| <i>Jun. 2021</i>             | Organiser, research group writing retreat at the Cass field station.                                                                                                                                                          |
| <i>Jul. 2019</i>             | Lab demonstrator, University of Canterbury Open Day.                                                                                                                                                                          |
| <i>May 2019 and 2023</i>     | Outreach activity, Hororata Night Glow Festival.                                                                                                                                                                              |

## KEY SKILLS

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- Atmospheric dynamics, extreme rainfall, and boundary layer meteorology.
- Field work and field data management, including radiosonde, lidar, and sonic-anemometers.
- Python programming: Adept at developing and maintaining scientific software for numerical models.
- Atmospheric numerical modelling: Experienced in WRF and PALM (Large Eddy Simulation).
- High-performance computing clusters: Compile models and conduct simulations (> 2,000 CPUs).
- Geospatial data analysis and visualisation: QGIS, ArcGIS and Python.
- Big data processing and analysis: Adept in handling and analysing large datasets (> 50 TB).